USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 22–23 LAB - Tapering, Squint, and Quantization
Lab Packet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 3.7: I can apply amplitude tapering (uniform, cosine, Chebyshev, Taylor) to control sidelobe level and predict the pattern trade-off.

Learning Objective 3.8: I can identify grating lobes, beam squint, and quantization effects in a phased array and describe their impact on system performance.

This packet is the L22–23 Lab, in three parts, and you turn it in once, complete, before Lesson 24. **Part 1, Predict**, is yours to do before the lab period: what each Part 2 table should read on your kit. **Part 2, Measure**, is your own kit's readings, recorded in class. **Part 3, Analyze**, is yours to write after the lab.

Grading. One submission, counted toward your engagement grade and graded as a complete, documented attempt. I grade each Part 3 answer on four things:

- a claim: what agrees, what does not, and why;
- the evidence: your own numbers from Part 2, set against your predictions;
- the reasoning that connects the evidence to the claim;
- the uncertainty: how large a difference your measurement can resolve.

Part 1: Predict (individual, before the lab period)

1. **Predict Tables A to E.** Fill in each table below with what the matching Part 2 table should read on your kit, at the HB100 frequency your kit reported with **Find HB100** in an earlier lab. If you did not record one, use the nominal 10.525 GHz and say so. The Pre-Lab frames of Lessons 22 and 23 name the frame that works each method, at 10.0 GHz.

   The array has $N = 8$ patches, E1 to E8, on a $d = 14$ mm pitch. Facing the patches, E1 is on the right and E8 on the left, and $+\theta$ is on the E8 side. The HB100 stands 1.2 m from the array. The Element Gains sliders are voltage amplitudes in percent of full scale, and the presets set them, from the end element inward and mirrored on the other half, to Hann 12, 43, 77, 100; Blackman 6, 27, 66, 100; and Chebyshev 4, 23, 62, 100. Table B is a symmetric taper of your own design, with an HPBW no wider than 16° and the first sidelobe below -20 dBc. Table C leaves Rx1, Rx4, and Rx7 on, then only Rx1 and Rx5. Table D puts the HB100 at 45° and sweeps at a **Signal BW** of 10 and then 500 MHz. Table E holds the beam at 15° with the Blackman taper and sweeps the HB100 in azimuth at 7, 4, 3, and 2 phase bits. Report levels as Part 2 does: lobe levels in dBc, Table C heights against its 0° lobe, and peak drops against the uniform trace.

Setup	Frequency	Far-field distance

Table A: preset	**HPBW**	**Peak drop**	**First sidelobe**
Uniform		reference	
Hann			
Blackman			
Chebyshev			

Table B: your design						
Rx1, Rx8	**Rx2, Rx7**	**Rx3, Rx6**	**Rx4, Rx5**	**HPBW**	**Peak drop**	**First sidelobe**

Table C: elements on	**Lobe angles**	**Height vs** 0°	**Peak drop**
Rx1, Rx4, Rx7			
Rx1, Rx5			

Table D: Signal BW	**Angle (Peak)**	**Shift**
10 MHz		reference
500 MHz		

Table E: bits	**Strongest lobe**	**Where**
7		
4		
3		
2		

Show your work for one row of each table.

Part 2: Measure (your kit, in class)

Follow the procedure on the L22–23 Lab page. Open the **Lab Presets** section before you press a preset. Press **Find HB100**, then **Calibrate**. Set **Rx Gain** so that the uniform main-lobe peak, source at boresight, reads -10 ± 3 dBFS on the FFT, and leave it there. Keep the source 1.2 m from the array, and keep $\pm 40^\circ$ of the kit clear of walls and people. Record each reading as you take it, and every level in dBc: decibels below the main-lobe peak of the same trace, its dBFS reading subtracted, so a level means the same thing on every trace.

Signal Freq =		Rx Gain =	
Distance =		Floor =	

2. **Table A: the presets.** Source at boresight, the Tapering preset, Enforce Symmetric Taper on. Freeze the uniform trace as the reference. If the uniform trace's two first sidelobes differ by more than about 2 dB, recalibrate before you go on.

Preset	HPBW	Peak drop	First sidelobe
Uniform		reference	
Hann			
Blackman (optional)			
Chebyshev (optional)			

3. **Table B: your design.** Load the taper you designed in Part 1, sweep, and adjust until it meets the specification: an HPBW no wider than 16° and the first sidelobe below -20 dBc. Record the final gains.

Rx1, Rx8	Rx2, Rx7	Rx3, Rx6	Rx4, Rx5	HPBW	Peak drop	First sidelobe

4. **Table C: grating lobes.** Source at boresight, Enforce Symmetric Taper off, the Grating Lobes preset (Rx1, Rx4, Rx7 on). Then Rx4 and Rx7 to 0% and Rx5 to 100%. Read each lobe's angle and its height against the 0° lobe, and the peak drop against the frozen uniform trace.

Elements on	d_{eff}	Lobe angles	Height vs 0°	Peak drop
Rx1, Rx4, Rx7	42 mm			
Rx1, Rx5	56 mm			

5. **Table D: beam squint.** Place the HB100 at about 45° on the 1.2 m arc and press the Uniform preset. Set **Signal BW** to 10 MHz, sweep, and record **Angle** (set to **Peak**). Then choose the Beam Squint preset, check that **Signal BW** reads 500 MHz (Calibrate, Find HB100, and the Rect preset all reset it to 0), and sweep again.

Signal BW	Angle (Peak)	Shift
10 MHz		reference
500 MHz		

6. **Table E: phase quantization.** The Quantization preset holds the beam at 15° with the Blackman taper and opens Signal vs Time. Sweep the HB100 in azimuth from -90° to $+90^\circ$ on the 1.2 m arc, in about 20 s, at Phase Shift Bits 7, 4, 3, and 2, with Use Bits checked before each sweep. On each trace read the strongest lobe outside the main lobe, in dBc, and where it is. Write "floor" where nothing rises out of it.

Bits	Strongest lobe outside the main lobe	Where
7		
4		
3		
2		

Before you leave: Use Bits on, Phase Shift Bits 7, Signal BW 10 MHz, and the Uniform preset.

Part 3: Analyze (individual, after the lab)

Answer each prompt from your own Part 2 readings and your own Part 1 predictions.

7. **The tapers.** Compare your taper measurements (Tables A and B) with your predictions, quantity by quantity. Build an error budget from your own readings that accounts for each disagreement. Then say which taper you would load on this kit and why, and what your peak drops don't explain about the cost of that taper.

8. **Grating lobes and squint.** Compare your grating-lobe run (Table C) and your squint run (Table D) with your predictions. Explain each difference that is larger than your measurement can resolve, and give the cause you think most likely. Say what you would change in either run to check that cause.

9. **Phase bits.** Compare your four quantization sweeps (Table E) with each other and with your predictions. From your data, state when rounding the phases matters on this array and when it does not, and what your four sweeps don't explain.

Documentation: _____